CO3 – AT1

Experiments-Based Assessment

SOURCE CODE:

```
#include <stdio.h>
void merge(int arr[], int l, int m, int r)
{
    int i, j, k;
    int n1 = m - l + 1;
    int n2 = r - m;
    int L[n1], R[n2];
    for(i = 0; i < n1; i++)
        L[i] = arr[l + i];
    for(j = 0; j < n2; j++)
        R[j] = arr[m + 1 + j];
    i = 0;
    j = 0;
    k = l;
    while(i < n1 && j < n2)
    {
        if(L[i] <= R[j])
            arr[k++] = L[i++];
        else
            arr[k++] = R[j++];
    }
    while(i < n1)
        arr[k++] = L[i++];
    while(j < n2)
```

```
        arr[k++] = R[j++];
}
void mergeSort(int arr[], int l, int r)
{
    if(l < r)
    {
        int m = (l + r) / 2;
        mergeSort(arr, l, m);
        mergeSort(arr, m + 1, r);
        merge(arr, l, m, r);
    }
}
int main()
{
    int arr[] = {45, 12, 89, 34, 67, 23, 90, 11};
    int n = sizeof(arr) / sizeof(arr[0]);
    printf("Original Array:\n");
    for(int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    mergeSort(arr, 0, n - 1);
    printf("\n\nSorted Array:\n");
    for(int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}
```